



Financial Vulnerability and Value Creation in Zero-Waste-Oriented Fish-Processing MSME: Evidence From a Pangasius Value Chain in Indonesia

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Abstract: This paper discusses the effects of the sustainable digital marketing approach on the green purchase intentions in the renewable energy market in Jordan and in particular the behavioral processes that mediate the formation of the environmentally responsible consumption. The quantitative research design was utilized to obtain data based on 237 consumers and processed with the help of the Partial Least Squares Structural Equation Modeling, to determine the direct, mediating, and moderating relationships. The results indicate that sustainable digital marketing initiatives have a significant positive impact on the environmental awareness and green purchase intentions. The environmental awareness can be identified as one of the main explanatory factors, and it mediates the connections between sustainable digital marketing and green purchase intentions, which shows that it is the key influential factor of pro-environment consumer behavior. In contrast, the moderating effect of perceived consumer effectiveness on the relationship between sustainable digital marketing strategies and green purchase intentions is found to be statistically insignificant. However, the Standardized Root Mean Square Residual (SRMR) values of the saturated and estimated models slightly exceeded the recommended 0.08 threshold, suggesting that the structural estimates should be interpreted with caution and that future research should validate the model using larger samples and additional fit indicators. These findings imply that the digital sustainability messages are effective in increasing awareness and behavioral intentions, but the individual perceptions of personal impact might not have a significant effect on strengthening this relationship in the context of the study. On the whole, the research finds that sustainability of digital marketing through the use of eco-friendly practices is a very crucial channel via which sustainable marketing strategies can predict green consumption behavior. The results advance the existing knowledge of the effects of sustainability-focused digital communication on consumer behavior in the developing markets and offer valuable practical recommendations to policymakers and marketers who may want to encourage the use of renewable energy by leveraging the behavioral-based digital communication approaches.

Keywords: Sustainable digital marketing; Green purchase intentions; Environmental awareness; Perceived consumer effectiveness; Renewable energy; Jordan

1. Introduction

The rapid expansion of aquaculture-based food systems has intensified sustainability challenges related to resource efficiency, waste generation, and economic resilience, particularly within micro, small, and medium enterprises (MSMEs). While aquaculture contributes significantly to food security, nutrition, and local economic development, processing activities frequently generate substantial by-products that remain underutilised. In many

small-scale processing contexts, these by-products are often treated as waste rather than as potential sources of additional value, which limits both environmental performance and long-term business sustainability.

Indonesia's extensive fisheries and aquaculture resources support food security, employment, and regional economic development. East Java is one of the country's strategically important fisheries regions, where aquaculture production and fish-processing activities provide opportunities for the development of value-added MSMEs (BPS Jawa Timur, 2024). However, the expansion of fish processing also increases the generation of skin, heads, bones, trimmings, and other processing residues. For small-scale processors, the limited utilisation of these materials may reduce resource efficiency and create additional operational and environmental burdens.

The growing demand for convenient and diversified fish products has heightened the importance of fish-processing MSMEs and the need to utilise raw materials efficiently. Fish-processing activities generate by-products that may pose economic and environmental burdens when not effectively utilised. Transforming these materials into marketable products is consistent with blue economy principles, which emphasise resource efficiency, waste minimisation, and value creation from aquatic resources.

The blue economy promotes the sustainable use of aquatic resources, resource efficiency, and the transformation of residual materials into useful inputs or economically valuable outputs while maintaining environmental integrity (Pauli, 2010; UNEP, 2018). This principle is consistent with the Zero Waste International Alliance's emphasis on responsible resource conservation, reuse, and recovery (ZWIA, 2018). Nevertheless, adopting blue economy principles at the MSME level does not automatically ensure business sustainability because by-product processing requires operational capacity, market access, regulatory compliance, and financial resources.

The increasing demand for processed fish products highlights the strategic role of MSMEs in the sustainability transition of the fisheries processing sector. Globally, MSMEs account for around 99% of all businesses, generate approximately 60% of employment, and contribute between 50% and 60% of value added, making them key drivers of sustainable and inclusive economic growth and critical agents in adapting to structural challenges such as digitalisation, globalisation, and environmental pressures (OECD, 2019). These conditions require MSMEs to continuously improve efficiency, product quality, and market responsiveness to achieve long-term business sustainability.

In Indonesia, MSMEs in the form of fish processing units (Unit Pengolahan Ikan—UPI) occupy a strategic position within the fisheries value chain, from processing to distribution, and are widely distributed across rural and remote areas. By producing value-added fish products, these enterprises contribute to employment creation, income generation, local economic development, and national food security. To strengthen this role, the Ministry of Marine Affairs and Fisheries (KKP) has initiated strategic programmes, including the revitalisation and upgrading of micro- and small-scale UPI, to support national priority agendas such as food self-sufficiency, downstream fisheries development, and the provision of nutritious food programmes.

Recent studies show that the transition of MSMEs towards circular and sustainable business practices depends not only on waste reduction but also on managerial capabilities, employee skills, technological resources, and market conditions (Dey et al., 2020; Gosh et al., 2022; Intyas & Primyastanto, 2020). In fisheries and agri-food processing, by-products can be transformed into commercially valuable outputs, thereby reducing environmental burdens and supporting circular resource use; however, their economic contribution depends on the feasibility of processing technologies and the existence of viable markets (Coppola et al., 2021). Business resilience also extends beyond short-term profitability because it reflects an enterprise's capacity to anticipate disruptions, cope with their immediate effects, and adapt its activities in response to changing conditions (Duchek, 2020). These capabilities are particularly important for MSMEs, whose limited financial and organisational resources may increase their exposure to supply, cost, and demand shocks (Eggers, 2020). Accordingly, financial resilience is understood as a dynamic capacity that develops over time, whereas financial vulnerability reflects the degree to which an enterprise's financial outcomes are exposed to adverse changes. Given that this study uses projected financial data rather than longitudinal observations of actual disruptions and adaptive responses, the sensitivity analysis assesses financial vulnerability rather than providing a comprehensive measurement of financial resilience.

Within this institutional and sectoral setting, a micro-scale UPI in Malang Regency represents the application of zero-waste-oriented pangasius processing. This configuration provides an empirical setting to examine whether zero-waste-oriented processing at the firm level enhances sustainability beyond resource efficiency. The UPI processes pangasius into fillets as its primary product and utilises usable processing materials through direct sale, internal processing, and other productive uses, thereby maximising raw-material utilisation and reducing solid material losses. The UPI continues to evolve through ongoing product development, including research on using fish oil from processing residues, while simultaneously strengthening its supply chain by collaborating with local aquaculture farmers to improve raw material continuity, upstream–downstream integration, and the overall sustainability of zero-waste-oriented fish processing at the local level.

The sustainability of fisheries-based MSMEs is closely linked to their ability to create and maintain a competitive advantage under sustainability constraints. One widely used analytical framework to examine competitive advantage at the firm level is Porter's value chain model. According to Porter (1998), the value chain framework conceptualises the firm as a configuration of interrelated primary and support activities through which

inputs are transformed into outputs that create value for customers. Competitive advantage arises not from the firm itself, but from how effectively these activities are configured and coordinated. Accordingly, value chain analysis serves as a strategic diagnostic tool to examine cost behaviour, differentiation potential, and margin formation, particularly in small-scale enterprises where operational activities strongly influence overall performance.

Previous research has applied value chain analysis to examine profitability, sustainability, governance, and upgrading opportunities in small-scale fisheries, showing that enterprise performance is shaped by activity configurations, market access, and institutional conditions (Rosales et al., 2017). Research on seafood-processing by-products further indicates that circular resource utilisation can reduce material losses and create additional value, although its economic viability depends on technological, market, and organisational conditions (Cooney et al., 2023). However, limited evidence explains how the internal configuration of activities in a micro-scale pangasius fillet-processing enterprise shapes its cost structure, financial outcomes, and vulnerability to adverse changes in costs and revenues. This gap is important because by-product utilisation may improve resource efficiency without necessarily strengthening the enterprise's capacity to withstand financial pressure.

In addition to operational considerations, business sustainability must also be assessed from an economic perspective. Financial feasibility analysis is essential not as a standalone investment appraisal but as a component of sustainability assessment to evaluate whether processing configurations and product diversification strategies generate sufficient economic returns to sustain business operations over time (Gittinger, 1982; Gray et al., 1992). Nevertheless, the utilisation of processing by-products may improve resource efficiency without necessarily strengthening the enterprise's ability to withstand financial pressure.

Accordingly, this study addresses the following research question: How does the configuration of primary and support activities in a zero-waste-oriented pangasius processing MSME influence its cost structure, economic feasibility, and financial vulnerability to changes in costs and revenues? Porter's value chain framework is applied as a strategic diagnostic tool to examine the configuration of internal activities, while financial analysis is subsequently applied to evaluate the profitability, investment feasibility, and financial vulnerability associated with that activity configuration.

Based on the above considerations, an integrated analysis combining value chain assessment and financial feasibility evaluation is required. Therefore, this study aims to analyse the sustainability of a pangasius catfish processing MSME in Malang Regency, Indonesia, by applying Porter's value chain framework alongside financial feasibility analysis to examine its internal activity configuration, evaluate its profitability and investment feasibility, and assess its financial vulnerability to increases in costs and decreases in revenues.

This study contributes to the sustainability literature by demonstrating that zero-waste-oriented processing at the MSME level does not automatically reduce financial vulnerability. While operational integration enhances resource efficiency, sustainability outcomes remain vulnerable under cost-driven value chain configurations. Conceptually, the study distinguishes activity-level value configuration from accounting-based financial returns. Empirically, it shows how operational concentration, limited market-oriented activities, and sensitivity to adverse financial changes jointly shape the sustainability of a small-scale fish-processing enterprise.

2. Research Method

This study adopts a descriptive research design with a case study approach to obtain an in-depth understanding of firm-level activities and business performance. According to Nazir (2003), a case study approach enables detailed examination of operational conditions, organisational characteristics, and managerial practices within their real-life context, allowing context-specific and comprehensive conclusions to be drawn.

Within this design, the selected enterprise is treated as a revelatory single case because it provides detailed access to an integrated pangasius fillet and by-product processing system, enabling an in-depth examination of the relationships among internal value chain activities, cost structure, revenue composition, and financial performance. The enterprise was selected based on three considerations: (1) it integrates pangasius fillet production with the utilisation of processing by-products within the same business system; (2) it maintains relationships with aquaculture farmers and uses different market channels for fillets and processed products; and (3) operational records, cost and revenue data, and information concerning its internal business activities were available for analysis. The analytical value of the case lies in its ability to provide detailed firm-level evidence that cannot be obtained from aggregate MSME data, particularly regarding how by-product utilisation is embedded within primary and support activities and how this configuration is associated with financial vulnerability. Therefore, the findings provide context-specific analytical insights rather than statistical generalisations to all Indonesian fish-processing MSMEs. The research was conducted at a micro-scale UPI in Malang Regency, Indonesia, that adopts a zero-waste-oriented processing approach for pangasius catfish.

The case also possesses several context-specific characteristics that may influence its value chain and financial outcomes. These include informal supply relationships with pangasius farmers, access to the government-supported Free Nutritious Meal (Makan Bergizi Gratis—MBG) programme as one of its market channels, an owner-centred management structure, recruitment through family and local community networks, and limited

storage capacity. Informal supplier relationships may support flexibility but do not guarantee long-term supply continuity, while access to the MBG programme may provide demand opportunities that are not available to all fish-processing MSMEs. The owner-centred structure and locally based workforce may reduce organisational complexity but may also limit formal managerial specialisation. These conditions should therefore be considered when interpreting the findings of this single case.

In this UPI, whole pangasius is processed into fillet products and several secondary outputs. According to Intyas et al. (2026), the skin, head, belly, and marketable trimmings are sold as secondary outputs. Residual flesh that does not meet the size or appearance specifications for fillet products but remains suitable for human consumption is processed into value-added products, particularly *fish sempo* and *tofu fish balls*. Fish bones are utilised as animal feed but are not recorded as a separate commercial product. The UPI is currently developing the further processing of belly portions into smoked fish. Based on the enterprise's processing standard, each whole pangasius is estimated to weigh approximately 500–800 g and produce approximately 280 g of unglazed fillet, consisting of two pieces. These figures represent approximate operational estimates because the weight of individual fish and processing outputs varies across production batches.

For the purposes of this study, “zero-waste-oriented processing” is operationally defined as the enterprise's efforts to minimise solid fish-processing waste through internal processing, direct sale, or other productive uses. Liquid waste generated from washing activities is managed in accordance with the applicable handling requirements and is not treated as a valorised output in this study. Based on the resource-conservation principles outlined by ZWIA (2018), this term describes the direction of the enterprise's material-utilisation practices rather than the complete elimination of every waste stream.

Data collection was conducted from May to December 2025 through direct observation, semi-structured interviews, questionnaires, and documentation. Intensive direct observation was carried out over a two-month period to examine daily production and processing activities, raw material utilisation, product diversification, by-product management, and the flow of materials from receipt to final product preparation. The enterprise employed 12 workers across its managerial, production, marketing, administrative, warehousing, and supporting functions. Four key informants, comprising the business owner and three employees, were purposively selected for having at least 5 years of involvement in the enterprise and direct knowledge of its operations and management. The main semi-structured interview with each informant lasted approximately one hour and focused on production processes, value chain activities, cost allocation, financial records, and business management. Additional questions and follow-up discussions were conducted alongside direct observation whenever further clarification was required. A structured questionnaire containing a standardised list of questions and data-recording fields was used to collect and verify raw production data, including input quantities, production volumes, product outputs, costs, asset ownership, and sales. Documentation was used to obtain supporting information from production, financial, and sales records, as well as other relevant enterprise documents. The information obtained was validated through methodological triangulation by comparing interview responses and questionnaire data with direct observations and enterprise records. Any inconsistencies were clarified through follow-up discussions with the relevant informants.

The data collected were analysed using Porter's value chain analysis and financial feasibility analysis. Value chain analysis was used to systematically map and classify internal business activities into primary and support activities, enabling the identification of activity configurations, cost behaviour, operational efficiency, and potential bottlenecks along the production process. A financial feasibility analysis was then conducted to assess the enterprise's economic viability by evaluating whether the existing processing configuration and product diversification strategies generate sufficient returns to sustain business operations over time.

The value chain results are presented in a descriptive–diagnostic manner to explain how individual activities contribute to value creation and cost efficiency. The value chain refers to a set of interrelated organisational activities through which firms create value for customers (Wilczek et al., 2024). Competitive advantage is shaped not only by external market conditions but also by internal organisational and managerial activities (Luo et al., 2024). Customer-perceived value generally emerges from activities that enable product differentiation, cost efficiency, and responsiveness to customer needs (David & David, 2017). Accordingly, firm profitability is closely linked to the effectiveness with which activities along the value chain are managed. Strategic focus on key activities can therefore support the achievement of sustainable competitive advantage (Ong et al., 2024; Viljoen et al., 2019). Porter's value chain framework, consisting of primary and support activities, provides a structured analytical lens for examining these relationships. Although Porter's framework was originally developed to examine competitive advantage, it is applicable to this MSME as a diagnostic tool for mapping internal activities, coordination, and cost concentration. However, its application is limited to an activity-level diagnosis and does not imply that the enterprise operates in a fully competitive market or possesses a formal competitive strategy.

Within Porter's framework, value activities are classified into primary and support activities. Primary activities include inbound logistics, operations, outbound logistics, marketing and sales, and service. Inbound logistics involve the receipt, storage, and handling of inputs, while operations encompass processes that transform inputs into final products, including the use and maintenance of production equipment. Outbound logistics involves

handling, storage, and distribution of products to buyers. Marketing and sales activities focus on promotion, pricing, and relationships with distribution channels, whereas service activities aim to maintain or enhance product value through after-sales support and customer assistance. Support activities consist of firm infrastructure, human resource management, technology development, and procurement, all of which facilitate and coordinate primary activities (Porter, 1998).

Porter's value chain framework was operationalised through three analytical stages. First, the enterprise's activities were identified from direct observations, interviews, questionnaires, and internal business records. Second, each activity was classified according to its principal function within Porter's framework. Activities directly involved in receiving input, transforming raw materials, distributing products, marketing outputs, and maintaining customer relationships were classified as primary activities, whereas management, human resources, technology development, and procurement were classified as support activities. Third, the preliminary classification was cross-checked against observational evidence, enterprise documents, and follow-up explanations from the key informants to ensure that it reflected the enterprise's actual operational structure.

Competitive advantage within the value chain may be achieved through cost leadership or differentiation strategies. In this study, cost efficiency and differentiation were examined at the activity level to identify strategic sources of advantage embedded within the firm's internal processes (Intyas et al., 2023; Kiplagat et al., 2025; Porter, 1998; Stabell & Fjeldstad, 1998). The value chain framework was further used to identify the structural sources of margin across activities. Margin, in this context, refers to the difference between the total value created by the configuration of activities and the cumulative cost of performing those activities. It is used as a diagnostic concept to explain where and how value is structurally generated along the chain and is empirically reflected in the realised profit generated by the enterprise. Profit and profitability were subsequently quantified using revenue and cost data, while financial feasibility was evaluated using investment and cash-flow indicators (Marimin et al., 2014; Porter, 1998).

The two analytical approaches were integrated sequentially at the interpretation stage while retaining their distinct analytical functions. Porter's value chain framework was first used to identify the configuration of primary and support activities, while the allocation of costs and assets was used to determine where financial resources and cost exposure were concentrated. This integration is consistent with value-flow approaches that combine material-flow and monetary information to identify resource losses, cost accumulation, and opportunities for circular value creation (Zhou et al., 2016; Zhou et al., 2017). Profitability and financial feasibility analyses were then conducted using the same underlying aggregate accounting and cash-flow data. The findings were jointly interpreted by examining whether the activity-level concentration of costs and assets was consistent with the enterprise's profitability, investment feasibility, and sensitivity to adverse changes in costs and revenues. This integration was interpretive rather than additive: value chain indicators were not treated as financial-return measures, and financial indicators were not used as substitutes for strategic value.

Financial feasibility analysis evaluates the enterprise's economic viability by examining its ability to generate returns relative to the capital invested and costs incurred. The analysis consists of three main components: profitability analysis, investment time-horizon assessment, and sensitivity analysis. Profitability indicators assess the efficiency of capital utilisation and income generation. Investment time-horizon analysis evaluates the period required to recover the initial investment and achieve acceptable returns. Sensitivity analysis examines how changes in key cost and revenue variables affect financial performance and feasibility. In this study, sensitivity analysis is used to diagnose financial vulnerability under specified adverse scenarios over the 10-year investment horizon. The projected cash flows are derived from baseline financial records and scenario assumptions. Although the analysis covers multiple projected periods, it does not capture observed adaptive responses to actual disruptions over time. A comprehensive evaluation of financial resilience would require longitudinal evidence of how the enterprise responds, adapts, and recovers from such disruptions.

Capital. Capital refers to production resources used to support further production activities, emphasising value, purchasing power, or use capacity embedded in capital goods. Capital may take the form of monetary resources or physical assets. Based on their duration of use, company assets are classified as either liquid or fixed. Liquid assets are consumed within a single production cycle and have a short turnover period (generally less than one year), whereas fixed assets are long-term assets that are consumed gradually during the production process and have a longer turnover period (Riyanto, 2010).

The investment appraisal adopted a 10-year planning horizon in accordance with the enterprise's long-term financial plan. Annual depreciation was calculated using the straight-line method by dividing the acquisition cost of each depreciable asset by its estimated useful life, while land was not depreciated. The estimated useful lives varied by asset category: 14–20 years for buildings, 10–15 years for vehicles, 3–10 years for freezing and cold-storage equipment, 5 years for major processing machinery, and 1–10 years for supporting equipment and utensils. Assets reaching the end of their useful lives within the planning horizon were replaced according to the relevant replacement schedule. The estimated replacement costs incorporated an annual asset-price escalation of 1%. Assets with unexpired useful lives at the end of year 10 were assigned proportional residual values, which were included as cash inflows in the final year. A discount rate of 6% was applied based on the estimated annual bank interest

rate adopted in the enterprise's 2025 financial plan. The same discount rate was used across all sensitivity scenarios to isolate the effects of changes in costs and revenues.

Cost. Costs are classified as fixed or variable. Fixed costs remain unchanged regardless of output level, while variable costs vary in proportion to changes in production volume (Chamdhani et al., 2025). Total cost is calculated as follows:

$$TC = FC + VC \quad (1)$$

where,

TC = Total Cost

FC = Fixed Cost

VC = Variable Cost

Accounting data were compiled from the enterprise's financial records to determine asset values and the components of fixed and variable costs. For the value chain analysis, relevant costs and assets were traced to the activities in which they were directly incurred or principally used. After allocation, the share of each activity was calculated as a percentage of total costs or total assets to describe the concentration of financial resources and cost exposure across the value chain. These percentages were not interpreted as direct measures of customer-perceived value or of the value created by each activity. At the aggregate level, cost, revenue, investment, and cash-flow data were used in the profitability and financial feasibility analyses.

Revenue. Revenue represents the total income generated by the business over a given period (Riyanto, 2010). Total revenue is calculated using the following formula:

$$TR = PQ \quad (2)$$

where,

TR = Total Revenue

P = Price

Q = Quantity

Profit. Profit is defined as the difference between total revenue and total costs incurred by the business (Agus & Nelwan, 2023). Profit is calculated as follows:

$$\pi = TR - TC \quad (3)$$

where,

π = Profit (Indonesian Rupiah—IDR)

Profit Margin. Profit margin expresses profit as a proportion of total revenue and is calculated as follows:

$$\text{Profit Margin (\%)} = \left(\frac{\pi}{TR} \right) \times 100 \quad (4)$$

Profit margin represents the proportion of total revenue remaining as profit after total costs have been deducted. It is interpreted alongside the value chain findings to describe the enterprise's economic performance, but it is not treated as a direct measure of the value created by individual activities.

Revenue Cost Ratio (R/C). R/C ratio compares total revenue with total costs to indicate operational feasibility; values greater than 1 indicate profitability, a value of 1 indicates break-even, and values less than 1 indicate a loss (Chamdhani et al., 2025), and is calculated as follows:

$$R/C \text{ Ratio} = \frac{TR}{TC} \quad (5)$$

Break-Even Point (BEP). BEP represents the level at which total revenue equals total costs (Gray et al., 1992), and is calculated as follows:

$$BEP_s = \frac{FC}{1 - \frac{VC}{TR}} \quad (6)$$

Return to Total Capital (RTC). RTC measures the profitability of total resources used in business operations. RTC value exceeding the interest rate indicates a stronger short-term capacity of the enterprise to recover invested capital (Gray et al., 1992). Business profitability is calculated as follows:

$$RTC = \frac{\pi - UFL}{M} \times 100\% \quad (7)$$

where,

UFL = Unpaid Family Labour

M = Working Capital (IDR)

Net Present Value (NPV). NPV measures the present value of net cash inflows minus the initial investment. An investment is considered feasible if $NPV > 0$ (Chamdhani et al., 2025). The NPV formula is as follows:

$$NPV = -A_0 + \sum_{t=1}^n \frac{A_t}{(1+i)^t} \quad (8)$$

where,

A_0 = Investment expenditure in year 0 (IDR)

A_t = Net cash inflows in year t (IDR)

i = Interest rate (%)

n = Economic life of the project (years)

Net Benefit Cost Ratio (Net B/C). Net B/C evaluates investment efficiency by comparing discounted benefits and costs; a value > 1 indicates that the investment is financially feasible (Chamdhani et al., 2025). Net B/C is calculated as follows:

$$\frac{B}{C} = \frac{\sum_{t=1}^n \frac{B_t}{(1+i)^t}}{\sum_{t=1}^n \frac{C_t}{(1+i)^t}} \quad (9)$$

where,

B_t = Gross income in year t (IDR)

C_t = Gross cost in year t (IDR)

Internal Rate of Return (IRR). IRR is the discount rate at which NPV equals zero; an investment is considered feasible when IRR exceeds the applied discount rate (Gray et al., 1992). IRR is calculated using the following interpolation formula:

$$IRR = i_1 + \frac{NPV_1}{NPV_1 - NPV_2} (i_2 - i_1) \quad (10)$$

where,

NPV_1 = Positive NPV (IDR)

NPV_2 = Negative NPV (IDR)

i_1 = Interest rate when NPV is positive (%)

i_2 = Interest rate when NPV is negative (%)

Payback Period (PP). PP represents the time required to recover the initial investment from annual net cash inflows (Gray et al., 1992). The PP is calculated as follows:

$$PP = \frac{\text{Initial Investment}}{\text{Annual Net Cash Flow}} \quad (11)$$

Sensitivity Analysis. Sensitivity analysis examines how changes in key variables affect financial performance and feasibility (Intyas et al., 2025). Three adverse scenarios were analysed: (i) a 15% increase in costs, (ii) a 12% decrease in benefits, and (iii) a simultaneous 9% increase in costs and 5% decrease in benefits. The resulting NPV, Net B/C, and IRR values were compared with the corresponding financial feasibility criteria.

3. Results and Discussion

Porter's value chain analysis is used to identify internal activities that drive value creation and margin formation within the firm. The framework explains how competitive advantage arises from the configuration of primary and support activities across the business. In this study, value chain analysis serves as a descriptive–diagnostic tool to examine patterns of value creation and cost structure, providing the basis for identifying operational efficiency, bottlenecks, and structural sources of margin prior to quantitative evaluation through financial feasibility analysis (Figure 1).

Figure 1 presents the enterprise's value chain, distinguishing the primary activities directly involved in product transformation and delivery from the support activities that facilitate their execution. The primary activity sequence begins with the receipt and inspection of pangasius, continues through fillet production and the internal transfer of processing by-products for further utilisation, and ends with product storage, distribution, marketing, and customer-related activities. Firm infrastructure, human resource management, technology development, procurement, and regulatory support operate across these primary activities. The figure provides the analytical

structure used to classify the enterprise's activities and subsequently map relevant costs and assets to their principal uses. It therefore serves as the link between the qualitative diagnosis of the enterprise's internal configuration and the subsequent interpretation of its cost structure and financial performance.

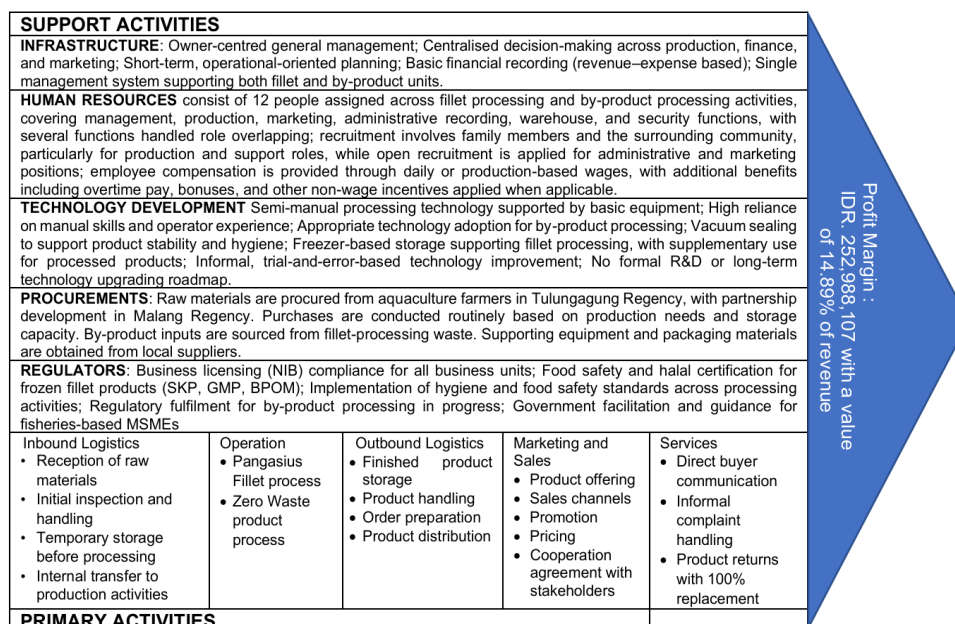


Figure 1. Structure of pangasius MSME's value chain in Malang, Indonesia

Note: NIB = Nomor Induk Berusaha (Business Identification Number); SKP = Sertifikat Kelayakan Pengolahan (Processing Feasibility Certificate); BPOM = Badan Pengawas Obat dan Makanan (Indonesian Food and Drug Authority); GMP = Good Manufacturing Practices; MSMEs = micro, small, and medium-sized enterprises; IDR = Indonesian Rupiah; R&D = research and development.

3.1 Value-Adding Primary Activities

Primary activities in the pangasius processing business consist of inbound logistics, operations, outbound logistics, marketing and sales, and services (Figure 1).

Inbound Logistics. Inbound logistics activities in the MSME are structured around the sequential flow of raw materials, from fillet processing to by-product processing. Fresh pangasius from aquaculture farmers is first received and undergoes an initial quality inspection, including a visual freshness assessment and basic sorting to separate raw materials suitable for fillet production. Incoming fish are weighed and recorded to support material control before entering the processing stage. Due to limited storage capacity, temporary holding relies on freezer units rather than large-scale cold storage facilities, which requires inbound logistics activities to be tightly synchronised with the fillet production schedule to minimise storage time and prevent quality deterioration. Following fillet processing, residual materials from this activity serve as internal inputs for by-product processing. These materials are not treated as external raw materials; instead, they are transferred directly within the production system to reduce handling time and storage requirements. This internal transfer mechanism shortens material flow, reduces waste accumulation, and ensures that by-products remain suitable for further processing. The inbound logistics system, therefore, functions not only to manage external raw material inflows for fillet production but also to coordinate the internal movement of materials from fillet outputs to by-product inputs, reflecting a simple yet integrated logistics structure adapted to the MSME's scale and capacity constraints.

Operations. Operational activities constitute the core transformation stage of the enterprise, covering both pangasius fillet production and the further utilisation of processing by-products. Operational activities begin with Pangasius fillet processing as the primary production process. Whole fish are cleaned and processed through head removal and evisceration, followed by the manual separation of flesh from bones and skin. The fillet meat is then trimmed to remove remaining impurities, rewashed, and drained before being weighed. After weighing, the fillets are packaged in food-grade materials and stored frozen prior to marketing and distribution. This process represents the main transformation stage in which raw fish are converted into the enterprise's core product.

Solid materials generated during fillet processing follow different utilisation pathways. The skin, head, belly, and marketable trimmings are sold as secondary outputs. Residual flesh that does not meet the size or appearance specifications for fillet products but remains suitable for human consumption is collected and processed internally into *fish sempol* and *tofu fish balls*. Fish bones are utilised as animal feed but are not recorded as a separate commercial product. The belly portion is currently sold, although its further processing into smoked fish remains

under development. These practices reduce solid material losses, but the degree of value addition and revenue contribution differs across the material categories.

Outbound Logistics. Outbound logistics activities are carried out after production and packaging are completed. Finished products, including frozen pangasius fillets and processed items such as *fish sempol* and *tofu fish balls*, are stored before distribution, with frozen products kept in freezers to preserve quality. During the storage period, products are handled in a simple manner to ensure that their condition and packaging remain intact. At this stage, distribution to external areas applies only to pangasius fillet products. Fillets are delivered directly by the MSME to several destinations, including Malang and Pasuruan, with a minimum order requirement to ensure efficient delivery. Processed products are distributed locally, as wider distribution is still under development.

Marketing and Sales. Marketing and sales activities focus on selling pangasius fillets and processed products once they are ready for delivery. Fillet products are marketed through distribution channels, including distributors and the MBG programme. In contrast, processed products such as *fish sempol* and *tofu fish balls* are sold directly to buyers. Sales channels, therefore, differ according to product type, reflecting their respective market characteristics.

Processed products are sold in limited quantities and mainly through direct sales, as broader market access remains constrained by ongoing licensing processes. Promotion activities remain limited and rely on existing business networks rather than formal marketing strategies. Fillet prices are determined through agreements with buyers, whereas processed-product prices are adjusted according to market conditions. Marketing and sales activities are further supported by cooperation with stakeholders, which facilitates product distribution and market access.

Services. Service activities play a very limited role in the value chain and do not yet contribute to systematic value enhancement. Although the enterprise provides direct buyer communication, complaint handling, product returns, and full product replacement when products do not meet the agreed quality requirements, these activities are conducted informally on a case-by-case basis and have not yet been formalised into a structured after-sales service system. Consequently, customer interactions remain reactive rather than proactive, reducing opportunities for product improvement and customer retention. This limitation also reduces the enterprise's ability to gather consistent market feedback to support product differentiation and pricing strategies. Therefore, service activities remain underutilised as a potential source of competitive advantage within the value chain.

3.2 Support Activities

Support activities are described at the top of the value chain and represent the institutional arrangements that enable the primary activities. Institutions and types of support include infrastructure, human resources, technology development, procurement, and regulators.

Firm Infrastructure. Firm infrastructure in the MSME is characterised by an owner-centred management system that supports both the fillet-processing and by-product-processing units. The business is managed under a centralised leadership structure, where the owner holds primary authority over managerial, financial, and organisational decisions. Business planning includes general management and financial management activities, such as raw material planning, resource utilisation, and coordination between the two processing units. However, the planning process remains largely flexible and informal, reflecting an operational rather than strategic orientation.

From an organisational perspective, the fillet processing unit applies a simple functional structure, in which key operational functions—such as production, quality control, marketing, and financial recording—are distributed among different individuals, resulting in overlapping roles under centralised supervision. In contrast, the by-product processing unit operates with a more informal, task-based organisational structure, without formal divisions, while remaining fully integrated within the same management system. Financial transactions are recorded using a basic manual system, in which income and expenditure are documented routinely and summarised weekly. This practice provides basic financial control but may also create a risk of recording inaccuracies.

Technology Development. Technology development in the MSME sector is primarily characterised by simple, semi-manual processing technologies that support daily production activities. The available production facilities enable processing activities—from raw material preparation to final product presentation—to be carried out efficiently and hygienically. Most production processes rely on basic equipment and operator skills, with technology functioning mainly as a supporting tool rather than as a fully automated system. In fillet processing, freezer-based storage is used to maintain product quality during temporary holding. In by-product processing, freezer use is supplementary and limited, while product stability relies more on processing techniques and packaging methods, such as vacuum sealing. Technological improvements are introduced gradually through practical experience and external assistance, without guidance from a formal research and development programme or a structured long-term technology upgrading plan. Regarding information technology, the by-product processing unit has received social media training through community service activities. However, social media use has not yet been consistently implemented in daily operations, and digital tools have not been integrated

into routine production management or marketing practices. As a result, technology adoption in this area remains at an introductory, non-operational stage.

Human Resource Management. Human resource management in the MSME supports both fillet processing and by-product processing activities under a single owner-centred management structure. The workforce is allocated across key functional areas, including management, production, marketing, administrative recording, warehouse, and security functions, which collectively serve the two processing units. Due to the small-scale nature of the enterprise, several functions are handled by the same individual, resulting in overlapping roles rather than a one-to-one correspondence between job functions and personnel. The workforce consists of permanent and daily/production-based workers, particularly in production activities related to both fillet and processed products. Recruitment is primarily conducted through family members and the surrounding community, especially for operational and support roles, while open recruitment is used selectively for administrative and marketing positions as needed. The owner is directly involved in selecting and supervising workers, and newly recruited workers undergo an informal adaptation period before fully assuming their responsibilities. Skill development is carried out internally through on-the-job training and direct instruction provided by senior workers and the business owner. Employee compensation is based on daily wages or production-based payments for non-permanent workers, complemented by overtime pay, bonuses, and other non-wage incentives when applicable, while permanent workers receive more regular wage arrangements. Basic workplace facilities are provided to support employee comfort and work continuity across both processing activities.

Procurement. Procurement activities in the MSME are centred on securing a continuous supply of raw materials and supporting inputs for integrated fillet and by-product processing. Raw materials for fillet production are primarily sourced from aquaculture farmers in Tulungagung Regency through routine purchasing and scheduled deliveries. However, raw material availability is occasionally constrained during off-season periods, creating supply challenges. In response, the MSME has initiated the development of procurement partnerships with aquaculture farmers in Malang Regency to strengthen supply resilience and reduce seasonal dependency. Supplier selection is largely based on established personal relationships and trust, resulting in non-contractual cooperation arrangements supported by mutual agreements on quality standards, pricing, and delivery continuity. The MSME provides suppliers with stable demand commitments through this informal partnership mechanism.

Procurement decisions are made with consideration of sales volume and existing storage capacity, although current storage limitations constrain the enterprise's ability to respond fully to rising demand. This condition is particularly relevant in the context of increasing market demand, including demand associated with government-supported intyasmes such as the MBG initiative. For by-product processing, raw materials are sourced internally from pangasius fillet-processing waste, enabling cost-efficient use of inputs and reinforcing the enterprise's zero-waste-oriented processing. Overall, procurement practices remain flexible, relationship-based, and adaptive, reflecting the operational characteristics and constraints of a small-scale MSME.

Regulators. Regulatory compliance in the MSME sector primarily involves meeting basic legal requirements, including holding a Business Identification Number (Nomor Induk Berusaha—NIB), which serves as the official legal identity for all business units and is a prerequisite for obtaining additional regulatory permits and government facilitation. For frozen fillet products, the MSME has obtained authorisation from the Indonesian Food and Drug Authority (Badan Pengawas Obat dan Makanan—BPOM) and holds a Processing Feasibility Certificate (Sertifikat Kelayakan Pengolahan—SKP), which confirms the implementation of Good Manufacturing Practices (GMP) and compliance with sanitation requirements. These measures ensure that the fillet products meet the hygiene, sanitation, and food safety standards required for wider market distribution. In addition, frozen fillet products have obtained halal certification, reinforcing consumer trust and compliance with mandatory halal regulations for food products in Indonesia. Across processing activities, the MSME applies food safety and hygiene practices in daily operations, including maintaining clean processing environments, handling raw materials properly, and controlling temperatures for frozen products. In contrast, regulatory compliance for by-product processing activities is still under development, with food safety authorisation and halal certification currently in progress as part of business expansion efforts. Regulatory implementation is further supported by facilitation, guidance, and training provided by fisheries- and MSME-related government institutions, which play a role in strengthening regulatory awareness, improving processing practices, and enhancing the legitimacy and sustainability of the enterprise.

At the value chain level, the analysis of primary and support activities indicates that the MSME's value chain configuration is largely cost-driven, with financial resources concentrated in internal operational activities and relatively limited market-oriented mechanisms for capturing value. Operational activities dominate the value chain, while marketing and service activities contribute relatively little to value capture, thereby constraining pricing flexibility and revenue stability under a cost-based pricing approach. The limited allocation of costs and assets to regulatory and other support activities further reflects reliance on external assistance from government and academic institutions. These findings suggest that improvements in value chain performance are more likely to result from better coordination and optimisation of existing activities than from structural expansion, with implications for long-term economic sustainability.

Building on this value chain configuration, financial analysis is conducted to translate activity-level

performance into measurable economic outcomes. Cost and revenue data are used to assess the cost structure, profit margin, and financial feasibility, with profit margin representing the immediate financial outcome of the value chain and financial feasibility evaluating whether returns are sufficient to justify the invested capital. The analysis begins with an examination of the capital structure, which comprises expenditures on processing equipment and supporting facilities and serves as the basis for fixed-cost and depreciation calculations, as presented in Table 1.

Table 1. Capital structure of the pangasius catfish processing MSME

No.	Description	Amount (IDR)
1	Land & buildings	415,000,000
2	Transportation assets	138,000,000
3	Cold storage facilities	170,000,000
4	Processing machinery	25,500,000
5	Supporting equipment & utensils	27,662,000
	Total	776,162,000

Note: IDR = Indonesian Rupiah; MSME = micro, small, and medium-sized enterprise.

Based on the asset values presented in Table 1 and their allocation across the value chain activities identified in Figure 1, the MSME's asset structure is largely concentrated in primary activities, with operations representing the most asset-intensive stage. Liquid assets are predominantly allocated to operational activities (43.27%), followed by outbound logistics (14.94%) and marketing and sales (2.89%), while inbound logistics and service activities each account for only 0.34% of total liquid assets. A similar distribution is observed for fixed assets, with operations again accounting for the largest share (27.56%), followed by outbound logistics (4.49%), marketing and sales (2.05%), and inbound logistics (1.00%). Only a small proportion of assets is assigned to service activities because complaint handling and product returns with full product replacement are conducted informally on a case-by-case basis and have not yet been formalised into a structured after-sales service system. Support activities also receive relatively limited allocations, including firm infrastructure (1.03%); human resource management, technological development, and procurement, each accounting for 0.69%; and regulatory compliance activities (1.00%).

From a Porterian value chain perspective, this pattern reflects a cost-driven value chain configuration, in which competitive advantage is primarily derived from operational efficiency rather than market-based differentiation. The strong concentration of both liquid and fixed assets in operations indicates that the enterprise's financial performance is highly dependent on processing efficiency, cold storage, and production handling. In contrast, the limited allocation of assets to marketing, service, and support activities suggests that these functions are not yet being strategically leveraged to create differentiation or improve customer retention. Consistent with Porter's theory, such a configuration supports short-term efficiency and cost leadership but may constrain long-term competitiveness unless greater emphasis is placed on strengthening market-oriented and support activities. This section examines the MSME's cost structure, classified as fixed and variable, as presented in Table 2.

Table 2. Annual cost structure of the pangasius catfish processing MSME

No.	Description	Amount (IDR)
Fixed Costs		
1	Depreciation	51,324,333
2	Permanent labour salaries	230,400,000
	Total Fixed Cost	281,724,333
Variable Costs		
1	Raw materials & food ingredients	910,062,000
2	Daily labour wages	108,000,000
3	Processing inputs & additives	21,408,000
4	Packaging & labelling materials	37,555,560
5	Utilities (electricity & water)	43,200,000
6	Distribution & logistics	30,600,000
7	Meals allowance & sanitation supplies	13,824,000
	Total Variable Cost	1,164,649,560
	Total Cost	1,446,373,893

Note: IDR = Indonesian Rupiah; MSME = micro, small, and medium-sized enterprise.

In terms of cost structure, Table 2 shows that the enterprise incurs total annual costs of IDR 1,446,373,893, comprising fixed costs of IDR 281,724,333 and variable costs of IDR 1,164,649,560. Variable costs account for approximately 80.52% of total costs, with raw materials and food ingredients representing the largest component. This structure indicates that financial performance is strongly influenced by production-related expenditure and

helps explain the enterprise's sensitivity to increases in operating costs.

Table 3 shows that the MSME's annual revenue is generated from several product categories, with different levels of contribution. Pangasius fillet products, particularly those with 50% glazing, account for the largest share of total revenue, indicating that income generation is strongly associated with the main processing output produced during the operations stage. Fillet glazing at 45% and 55% also makes a substantial contribution, confirming that variations in fillet specifications remain an important source of revenue.

Table 3. Annual revenue of the pangasius catfish processing MSME

No.	Product	Sales Quantity (kg/Year)	Amount (IDR)
1	Pangasius fillet (glazing 45%)	775	279,000,000
2	Pangasius fillet (glazing 50%)	2,445	821,520,000
3	Pangasius fillet (glazing 55%)	480	149,760,000
4	Fish skin	176	16,896,000
5	Fish head	1,804	16,236,000
6	Fish belly	264	17,424,000
7	Trimming residues	308	11,088,000
8	Premium trimmings	88	8,448,000
9	<i>Tofu fish balls</i>	2,040	195,840,000
10	<i>Fish sempo</i>	2,035	183,150,000
Total (IDR)			1,699,362,000

Note: IDR = Indonesian Rupiah; MSME = micro, small, and medium-sized enterprise.

Processed products derived from further utilisation of raw materials contribute a notable share of annual revenue. These products generate additional income through secondary processing activities, demonstrating the role of product diversification in revenue generation. In contrast, revenue from by-products, including fish skin, head, belly, and trimmings, accounts for a smaller share of total revenue, reflecting their complementary role to the main processed outputs. The utilisation of these materials provides two conceptually distinct economic benefits. The sale of skin, head, belly, and marketable trimmings generates supplementary revenue, while the internal processing of residual edible flesh and the utilisation of fish bones as animal feed may reduce the amount of material requiring disposal. However, this study did not quantify the counterfactual costs of disposal, transportation, or environmental compliance that would have been incurred without these practices. Therefore, the avoided-cost contribution was not included in the financial indicators. Accordingly, the utilisation of processing by-products is interpreted as evidence of improved resource use and supplementary revenue generation, but not as evidence of a substantial direct contribution to financial value creation.

Based on the revenue structure described in Table 3, the analysis is subsequently extended to evaluate the MSME's profitability and financial feasibility. Revenue data are combined with the total cost structure presented in Table 2 to assess whether the realised income is sufficient to cover production and operating costs and to generate a positive return. The results of the profitability and financial feasibility analysis, including key indicators for evaluating business performance, are presented in Table 4.

Table 4. Profitability and financial feasibility of the pangasius catfish processing MSME

Analysis	Unit	Amount
Profitability Analysis		
BEP sales	IDR	895,344,097
Profit	IDR	252,988,107
Profit margin	%	14.89
R/C	—	1.17
RTC	%	14.84
Financial Feasibility (Discount Rate 6%, 10-Year Period)		
NPV	IDR	1,411,085,950
Net B/C	—	2.82
IRR	%	37
PP	Years	2.70

Note: BEP = Break-Even Point; R/C = Revenue-Cost Ratio; RTC = Return to Total Capital; NPV = Net Present Value; Net B/C = Net Benefit-Cost Ratio; IRR = Internal Rate of Return; PP = Payback Period; IDR = Indonesian Rupiah; MSME = micro, small, and medium-sized enterprise. The symbol “—” indicates no unit.

Based on the results presented in Table 4, the pangasius catfish processing MSME is profitable and financially feasible. The business reaches a break-even sales volume of IDR 895,344,097 and generates an annual profit of IDR 252,988,107, indicating that total revenue exceeds total operating costs. The enterprise also records a profit margin of 14.89%, representing the proportion of annual revenue remaining after total costs have been deducted.

This financial outcome corresponds to the realised margin presented in the value chain configuration in Figure 1. This condition is supported by an R/C ratio of 1.17 (>1) and a RTC of 14.84%, which is higher than the prevailing interest rate of 6%.

The financial feasibility analysis at a 6% discount rate over 10 years shows a positive NPV of IDR 1,411,085,950 and a Net B/C of 2.82, confirming that discounted benefits outweigh discounted costs. The IRR of 37%, which exceeds the discount rate, and a PP of 2.70 years indicate that the investment can recover its initial capital within a relatively short time frame. Despite the favourable baseline results, the enterprise's financial feasibility may still be affected by adverse changes in costs and revenues. Therefore, a sensitivity analysis was conducted to evaluate its financial vulnerability under the scenarios presented in Table 5.

Table 5. Sensitivity analysis of the pangasius catfish processing MSME

No.	Assumptions	(%)	NPV (IDR)	Net B/C	IRR (%)
1.	Increased costs	15	-129,066,980	0.83	2.5
2.	Decreased benefits	12	-89,808,320	0.88	3.6
3.	Increased costs	9	-138,378,420	0.82	2.0
	Decreased benefits	5			

Note: NPV = Net Present Value; Net B/C = Net Benefit-Cost Ratio; IRR = Internal Rate of Return; IDR = Indonesian Rupiah; MSME = micro, small, and medium-sized enterprise.

Sensitivity analysis was conducted to evaluate the robustness of the business under changes in costs and benefits (Oraee et al., 2011). As shown in Table 5, the results indicate that the financial feasibility of the pangasius catfish processing MSME is sensitive to adverse changes in key variables.

A 15% increase in costs results in a negative NPV of IDR -129,066,980, a Net B/C below 1 (0.83), and an IRR of 2.5%, indicating financial infeasibility. Similarly, a 12% decrease in benefits leads to a negative NPV of IDR -89,808,320, a Net B/C of 0.88, and an IRR of 3.6%. When cost increases (9%) and benefit decreases (5%) occur simultaneously, the business also becomes financially infeasible, with an NPV of IDR -138,378,420, a Net B/C of 0.82, and an IRR of 2.0%. These results suggest that the business has limited tolerance for cost increases and revenue declines.

When interpreted jointly, the value chain and financial results explain the structural sources of the enterprise's financial vulnerability. The concentration of costs and assets in operations means that the enterprise's financial performance depends heavily on raw-material utilisation, processing efficiency, labour requirements, utilities, and production continuity. This exposure is reflected in the sensitivity analysis, in which a 15% increase in costs rendered the investment financially infeasible. On the revenue side, pangasius fillets generated approximately 73.57% of total annual revenue, indicating substantial dependence on the primary product despite the utilisation of by-products. This concentration helps explain why a 12% decline in benefits also produced an infeasible financial outcome. Therefore, the integrated interpretation does not suggest that high operational expenditure represents high value creation; rather, it shows that the enterprise's current value chain is financially dependent on cost-intensive operations and stable fillet revenue. By-product utilisation broadens the product portfolio and provides supplementary income, but it does not fully offset the financial exposure arising from operational cost concentration and primary-product dependence.

These findings extend existing research by showing how financial vulnerability is associated with the internal configuration of activities in a micro-scale fish-processing enterprise. Previous value chain research in small-scale fisheries has emphasised the roles of activity coordination, market access, governance, and upgrading in shaping enterprise outcomes (Rosales et al., 2017). Research on seafood by-product valorisation has also shown that improved material utilisation can create economic opportunities, although its viability depends on technological, market, and organisational factors (Cooney et al., 2023). The present case adds evidence from an enterprise with a narrower revenue structure centred on pangasius fillets. These findings reinforce the theoretical distinction between material circularity and financial sustainability: by-product utilisation can improve resource efficiency, but its financial contribution depends on cost control and market development. Accordingly, managerial and public support should extend beyond production equipment and waste reduction to include product-level cost monitoring, licensing and certification, market access, and the commercial development of products derived from processing by-products.

4. Conclusions

This study provides an integrated assessment of how the sustainability of a pangasius-processing MSME implementing zero-waste-oriented processing is shaped by its internal value chain configuration and associated financial outcomes. The analysis shows that operational activities account for the largest share of asset allocation and total costs. This concentration does not necessarily indicate that operations constitute the primary source of value creation; rather, it shows that the enterprise's financial performance is highly dependent on raw-material

costs, processing efficiency, labour, utilities, and production continuity. The enterprise's zero-waste-oriented processing supports resource utilisation through the sale of skin, head, belly, and marketable trimmings, the processing of residual edible flesh into *fish sempol* and *tofu fish balls*, and the utilisation of fish bones as animal feed. These practices reduce solid material losses and provide supplementary economic benefits, although they do not eliminate every waste stream. While the baseline analysis demonstrates economic viability, the vulnerability analysis should not be equated with a comprehensive assessment of financial resilience, which would require longitudinal evidence of adaptive responses to actual disruptions.

Under the baseline assumptions, the enterprise is financially feasible. However, the sensitivity analysis indicates financial vulnerability to increases in costs and decreases in revenues, particularly because the value chain remains concentrated in cost-intensive operational activities and primary-product revenue. Nevertheless, the vulnerability analysis provides an initial diagnosis of the financial pressures that may constrain such resilience. Therefore, long-term sustainability requires not only improved material utilisation but also stronger product-level cost monitoring, marketing capabilities, licensing and certification, market access, and the further development of value-added products. As the evidence is derived from a single enterprise, the findings should be interpreted as context-specific analytical insights rather than generalisable conclusions for all fisheries-processing MSMEs.

Future research should compare fish-processing MSMEs with different supply arrangements, market channels, organisational structures, and levels of access to government programmes to assess the transferability of these findings across different contexts.

Author Contributions

Conceptualization, C.A.I. and A.T.; methodology, C.A.I. and A.T.; validation, A.T. and E.C.A.; formal analysis, C.A.I., S.E.S., and A.K.; investigation, C.A.I., S.E.S., and A.K.; resources, C.A.I.; data curation, C.A.I., S.E.S., and A.K.; writing—original draft preparation, C.A.I.; writing—review and editing, A.T. and E.C.A.; visualization, S.E.S. and A.K.; supervision, A.T.; project administration, C.A.I.; funding acquisition, C.A.I. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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